

SUPPLEMENT TO “NEURAL BELLMAN OPERATORS”

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This supplement gives the primitive-domain verification, numerical and statistical details, complete resource records and direct economic comparisons for the retained R49 evidence and the R51 revision. The reference-advantage, terminal-improvement and finite-sweep accuracy proofs are included here; all earlier proofs remain in the complete development supplement. The records distinguish deterministic mathematical guarantees, outward numerical enclosures and the independent-bin assumption supporting finite-sample inference.

1. PRIMITIVE DOMAIN AND EXACT MODULI

Use the family in main equations (37)–(38), with $d = 2, 3, 4$. Capacity is between $1/8$ and $1/4$ and has ℓ^1 modulus $1/(8d)$. Every component of the dynamics is at least $1/32$. A direct upper bound is $29/32$ for the larger action exposure and $27/32$ for the smaller one. Thus the cube is invariant without an action-dependent exit event. The numerical clipping to $[0, 1]$ is inactive on the true primitive image; it does not authorize out-of-domain evaluation.

For $d \geq 2$, the sums of the absolute entries in a state column of the transition derivative are bounded by $1/2 + 1/16 + 1/8 = 11/16$. The action derivative has ℓ^1 norm $\sum_j \gamma_j$. The action derivative of current cost is $2pa + 16a^3 \leq p/2 + 1/4$. These are bounds between feasible pairs, not a claim that the same action is feasible at both states.

On either shortage region the state gradient is affine. The derivative of the squared-target and imbalance terms in coordinate j , after multiplication by d , is $4(x_j - 5/8) + (2x_j - x_{j-1} - x_{j+1})/2$. In the shortage region add $-8(1/2 - 2\sum x/d)$. The terminal target derivative replaces 4 by 8. A linear functional on each region reaches its extrema at region vertices. For the declared dimensions, these are cube vertices and vertices with all but at most one coordinate at zero or one and $\sum x = d/4$. Exact evaluation gives absolute derivative bounds $13/(2d)$ and $9/d$. The added

regression verifies these vertices rationally. The proof does not assert that these convenient formulas remain valid for untested arbitrary d .

The state covering radius of a nonuniform tensor grid is half the sum of its maximum coordinate spacings. Each action net is generated from a downward enclosure of capacity. Its coverage is at most $1/(8K) + 2^{-48}$ under the implemented binary64 construction. This last conservative allowance covers capacity enclosure, multiplication, division and the final downward displacement on the bounded action domain. Fixed spacing and valid interval arithmetic are checked; a negative action is clipped to zero. Downward quantization on deployment preserves feasibility because the capacity itself is enclosed from below over the complete acquisition cell.

2. NUMERICAL LABELS AND THE SEPARATED ACCOUNT

An objective query has lower and upper endpoints for its midpoint-sum expectation, plus the deterministic continuous-law integration remainder. Labels are rounded on the inherited dyadic lattice, and the largest distance to the objective endpoints supplies the query allowance e_t . Selecting the least rounded label and retaining its original action costs one allowance in the label comparison and another in the complete selected-policy account. It does not require a second copy of the action-cover term in main equation (7).

For a uniform scalar bin of width w , the mean absolute distance to its midpoint is $w/4$. Here $w = 1/(16M)$ and the transition's shock modulus is d . Therefore the numerical account decomposes into $\beta d L_{t+1}/(64M)$ and the remaining arithmetic width. The recorded primitive coefficients in main equation (24) reconstruct each witness bound as $a/N + b/K + q/M + \eta$ with a nonnegative numerical remainder. The release checks this decomposition at all witness rungs, as well as the full signed formula for every conventional rung.

The continuous allocation in main Proposition 11 minimizes a leading product, not a full finite-precision program. With positive coefficient c_i and exponent p_i , changing variables to error contribution $x_i = c_i/r_i$ gives the objective $\sum_i p_i \log x_i$. Its constrained first-order condition is $x_i = \min(\bar{x}_i, p_i \lambda)$. The multiplier exists uniquely until every cap binds. A zero coefficient has no benefit from refinement, so its coordinate stays at the resource lower bound. Upward dyadic rounding keeps every

error term weakly smaller. Compiler setup, integer counts, coefficient bit lengths, adaptive pilots, memory and durable output are not removed by that relaxation.

3. WHY THE COMPILED ACTOR IS THE ORIGINAL ACTOR

At a tensor node the distance transform minimizes over original sites, not merely neighboring transformed nodes. Each forward relaxation along a coordinate line compares the current value–owner pair with the preceding pair plus the weighted coordinate distance; the backward pass does the same from the opposite direction. The invariant is the lexicographic minimum over all processed original sites in the processed coordinates. Separability proves the invariant after the final coordinate. Keeping only the winning value would lose the actor identity.

For an off-grid point, choose a cell corner coordinatewise between the point and its lexicographically first minimizing site. Distances add exactly. At that corner an owner giving a smaller value, or a tied value and smaller original index, contradicts the minimizer at the original point. This proves the owner identity even for inconsistent labels, zero slope, boundary cells and ties. Nonuniform coordinate spacings enter only the exact distances.

In interval execution, value expressions are enclosed outward. Actor candidates whose lower objective exceeds another candidate’s upper objective can be discarded; the remaining possibilities must be retained. An exact point tie is resolved by rational comparison and the original index rule. A genuine uncertain state box is not an exact point: the hull of its surviving actions is propagated. Boxes crossing cells use the union of all relevant corners. These operations may cost more than one point query, and the direct records retain actor and measurement ambiguity counts.

4. ACQUISITION FOR WITNESS AND CONVENTIONAL POLICIES

Let $x \in C$ be the true state and $\hat{x}$ the reported midpoint, with radius r in ℓ^1 . Fix the selected original node x_i and node action a_i . The ideal repair at the true state is feasible and within $K^A \|x - x_i\|_1$ of a_i . Replacing that repair by the cell-wide lower-capacity repair followed by downward spacing Δ changes the action by at most $2K^A r + \Delta$. Thus

$$Q(x, a^{\text{impl}}) \leq y_i + e + L^{\text{gr}} \|x - x_i\|_1 + D(2K^A r + \Delta).$$

For a witness selection at $\hat{x}$, $y_i + L^{gr}\|\hat{x} - x_i\|_1 = f(\hat{x})$, up to a separately certified selector error. The triangle inequality and $L^f = L^{gr}$ yield the allowance in main equation (9).

For conventional nearest-node selection, subtract and add $f(\hat{x})$. The expression $y_i + L^{gr}\|\hat{x} - x_i\|_1 - f(\hat{x})$ is bounded by its complete nearest-cell maximum E_t . The remaining terms are at most $(L^{gr} + L^f + 2DK^A)r + D\Delta$. This proves main equation (34) without a continuity assumption on the actor. It also explains why applying the witness coefficient unchanged to an interpolated conventional critic would be incorrect when the actual critic modulus differs.

For two-state records, $r(b) = 2^{-b}$ and the common spacing is $1/4096$. The all-state bit criterion is therefore $G + K_\Delta + A2^{-b} + 2\lambda b \sum_{t < T} \beta^t$, where A uses the appropriate actual critic modulus and $K_\Delta = \sum_t \beta^t D_t / 4096$. The interval-net criterion uses actual expected costs on the same finite grid instead of replacing $G + K_\Delta + A2^{-b}$ by a claim about true loss.

5. CONTINUOUS PATH LAW AND SIMULTANEOUS INFERENCE

Conditional on all construction records, the studied controllers are fixed. Each simulation row draws independent uniform indices for the initial coordinates and each innovation. The interval simulator encloses the complete continuous cell corresponding to those indices; no bin midpoint is substituted for the true path distribution in the estimand. Uniformly mixing conditional continuous cells reproduces the original uniform initial and innovation laws. Common innovations may couple two policies, but each follows its own endogenous trajectory.

In dimension two, a valid support interval for a raw discounted path-cost difference is $[-H, H]$, where

$$H = \frac{99 + 4p}{64} \sum_{t < T} \beta^t + \frac{37}{16} \beta^T.$$

Indeed, current cost is nonnegative and at most $(99 + 4p)/64$ by separately bounding squared-target, imbalance, shortage and action costs. Terminal cost is at most $37/16$. These conservative bounds apply to every feasible repaired path. Clipping interval endpoints to the support preserves inclusion. Lower endpoint expectations lie below the true cost difference; upper endpoint expectations lie above it.

The stored moments consist of outward sample-mean intervals and upper sample-
variance bounds for each endpoint sample. In (29), the square root is rounded
upward after checking its square against the rational radicand. The linear remain-
der is rational. With $K \leq 300$, $\alpha = 1/100$ and $\ell = 13$, a finite positive Taylor sum
verifies $e^{13} > 4K/\alpha$. The bounded empirical-Bernstein inequalities of Maurer and
Pontil (2009) and a union bound therefore yield the claimed simultaneous event.
Dependence between contrasts through shared economic objects is permitted; in-
dependence within each bin-index sample is the statistical assumption. The fixed
pseudorandom seeds do not prove that assumption.

A first-crossing choice is measurable with respect to construction information.
The conditional coverage assertion consequently remains valid after this choice.
A bit selector or a fee queried after seeing the simultaneous intervals also needs
no additional error allocation: on the one event, each underlying cost difference
already belongs to its interval. This does not validate new initial laws, new trained
policies, or unrecorded contrasts. Old and new study families have separate error
accounts; combining them incurs the corresponding union bound.

6. EXACT FRONTIERS AND DECISION SEMANTICS

For a fixed construction ladder, the first-crossing function can change only at a
recorded certificate. A sorted union of all such values, together with zero, partitions
all positive tolerances into left-closed, right-open intervals. At an attained endpoint
the weak inequality in the stopping rule applies. Later nonmonotone or dominated
certificates need not change the function, but remain in the raw record. Adjacent
intervals with identical returned rungs may be merged for display; the exact rational
partition is deposited.

For a direct interval $[l, u]$, the gain from replacing witness by FVI is in $[l, u]$;
the reverse gain is in $[-u, -l]$. The fee $\max(0, u, -l)$ suffices to certify that neither
switch is strictly profitable. The smaller $\max(0, l, -u)$ only says that neither switch
has an interval lying wholly in its strictly profitable region. When zero lies inside
a wide interval, this latter threshold is zero and is not evidence that a free switch
cannot pay. The release reports both objects.

For information price λ , each bit choice supplies lower and upper affine net-cost lines. Enumerating all their nonnegative intersections partitions the price half-line. On an open interval, the active lower and upper lines are fixed. At a boundary all lines are compared exactly with smallest-bit ties. The difference between the active upper and lower lines is the selected implementation's valid regret upper bound. The identity line at 16 bits is exactly zero before information charges. No simulation count or time is assigned to an unperformed identity experiment; its arithmetic identity is recorded explicitly.

7. REFERENCE EVALUATION AND SAFE IMPROVEMENT

We give details of the reference-policy argument to separate a true improvement theorem from an approximate-greedy heuristic. All comparisons are on the same invariant domain and common conditional-evaluation domain as the main article. Feasible measurable policies include acquisition, capacity repair and numerical action realization.

Let $e_t = J_t^\pi - h_t$. At the terminal date $e_T \in [a_T, b_T]$. If $e_{t+1} \in [a_{t+1}, b_{t+1}]$, cash invariance and monotonicity yield

$$\beta_t a_{t+1} \leq \mathcal{T}_t^\pi J_{t+1}^\pi - \mathcal{T}_t^\pi h_{t+1} \leq \beta_t b_{t+1}.$$

Adding the signed own-policy residual proves the backward enclosure in Proposition 4. Applying the same statement with either feasible action proves the difference allowance. No independence or differentiability of the fitted critic is required. The domain hypothesis matters: a certainty equivalent need not satisfy the argument for an inadmissible translated continuation.

If $\tilde{h}_t = h_t + k_t$, then $\tilde{e}_t = e_t - k_t$ and the residual shifts by $\beta_t k_{t+1} - k_t$. The propagated intervals become $[a_t - k_t, b_t - k_t]$, so their widths are unchanged. The two action evaluations of $\tilde{h}_{t+1}$ each shift by $\beta_t k_{t+1}$, leaving the contrast unchanged. This is the centered invariance of the improvement test.

For the full expectation identity, define $A_t^\pi(x, a) = Q_t(J_{t+1}^\pi; x, a) - J_t^\pi(x)$. Along any feasible μ trajectory,

$$\mathbb{E}_x^\mu \sum_{t < T} B_t A_t^\pi(X_t, \mu_t(X_t)) = \mathbb{E}_x^\mu \sum_{t < T} \{B_t c_t(X_t, \mu_t(X_t)) + B_{t+1} J_{t+1}^\pi(X_{t+1}) - B_t J_t^\pi(X_t)\}$$

$$= J_0^\mu(x) - J_0^\pi(x).$$

Boundedness and finite horizon justify expectation and summation. Rejected actions give exactly zero reference advantage, while accepted ones give at most $-\kappa_t$. This proves Theorem 5. Under a nonlinear conditional recursion, use backward monotonicity instead of this telescoping expectation identity. A nonlinear evaluation does not justify the occupancy formula by changing notation alone.

The state-cell version fixes an action and verifies its inequality uniformly on the whole cell. Testing a cell midpoint alone is insufficient. Every implemented true state belongs to a cell whose gate is valid, and boundary cells use a declared deterministic report rule. In interval simulation, boxes straddling report boundaries retain all eligible actions, independently of whether their point-state selection is discontinuous. The analytic nonincrease may tighten a numerical advantage enclosure by intersection with $(-\infty, 0]$, because the support is established independently of the simulated sample. This is not truncation of an unfavorable sample estimate.

8. EXACT INTEGRATION AND ACTION OPTIMIZATION

Write $Z \sim U[-1/32, 1/32]$. The deterministic part of each next coordinate is y_j , and its random part is $s_j Z$. Hence the expectation of $(y_j + s_j Z - c)^2$ is $(y_j - c)^2 + 1/3072$. For the cyclic difference use coefficient $s_j - s_{j+1}$. Summing yields the quadratic and variance terms in equation (15).

For the shortage term, the affine random component is uniform on $[-r, r]$, where $r = |\sum_j s_j| / (16d)$. Direct antiderivatives give, for $r > 0$,

$$\mathbb{E}(b + U)_+^2 = \frac{(b + r)_+^3 - (b - r)_+^3}{6r}, \quad \mathbb{E}(b + U)_+ = \frac{(b + r)_+^2 - (b - r)_+^2}{4r}.$$

For $r = 0$ take b_+^2 and b_+ . These formulas hold at the two boundaries as well as inside the crossing region. The implementation encloses cancellation outward. Independent regression checks split the innovation interval at the exact shortage root and integrate the resulting quadratic pieces by exact rational Simpson identities; the checks are not used to generate the study's continuation labels.

At fixed x , the transition is affine in a . The terminal objective is a sum of convex quadratics and squared positive parts of affine functions. Its expectation is convex.

The action cost $pa^2 + 4a^4$ has second derivative $2p + 48a^2 \geq 2p$. Thus final conditional cost has a unique continuous minimizer, possibly on the capacity boundary. Its lattice minimizer lies at the first lattice point with nonnegative derivative or the preceding point, with boundary truncation. The implementation searches at most the fixed 4096 denominator's range, compares the two costs, and uses exact rational comparison when outward point intervals fail to decide a tie. The incumbent need not be one of these two actions: safety is established by the subsequent whole-cell comparison.

For bits b , let integer cell indices be $k_1, \dots, k_d$. The smallest capacity on the cell is $1/8 + \sum_j k_j / (8d2^b)$. Its largest lattice index is exactly

$$\left\lfloor \frac{4096(d2^b + \sum_j k_j)}{8d2^b} \right\rfloor.$$

Integer arithmetic avoids an upward feasibility error, including in dimension three. The original action is clipped and rounded downward by this same rule. Feasibility is therefore not inferred from an interval midpoint. The final gate keeps the same earlier policies; its true improvement is relative to that acquired incumbent, not to a different exact-state policy.

9. SIMULTANEOUS ACTUAL-COST INFERENCE

The R50 primary family contains 25 pairs of baseline generators. Each group contains four policies—the two incumbents and their final repairs—and produces four absolute costs and six pairwise differences. Thus there are 250 sampled estimands. Their selection uses frozen construction information only. Within each group the continuous initial law and the first $T - 1$ innovations are represented by forty-bit bins with outward propagation. The last innovation is integrated analytically. Conditional expectation over this last innovation preserves each policy's expected total cost. It reduces simulation variance without substituting a new innovation law.

The path-cost support derived in the retained inference section applies in dimensions two through four: the target and imbalance terms are coordinate averages with the same bounds, the shortage term is bounded by $1/2$, and feasible actions

do not exceed $1/4$. Consequently $0 \leq J\text{-score} \leq H$, with H as in Section 5; a paired score lies in $[-H, H]$. An incumbent-minus-own-repair contrast is nonnegative by the independently proved gate and is bounded by H . Because the earlier trajectory is common, its numerical enclosure uses the centered final contrast instead of subtracting two complete total-cost intervals.

For endpoint observations $X_i \in [A, B]$, the retained empirical-Bernstein bound uses

$$R_n = \sqrt{2\text{Var}(X)\ell/n + 7(B - A)\ell/[3(n - 1)]},$$

with outward moment intervals, an upper sample variance, and a rationally checked upward square root. Lower endpoints and upper endpoints each receive a one-sided bound. Set $n = 131072$, $\ell = 13$ and $\alpha = 1/100$. The family cap is 512; an exact finite Taylor sum proves $e^{13} > 4(512)/\alpha$. A union bound therefore covers all 250 actual-cost estimands under the independent-bin model. The cap leaves room but does not certify unrecorded models or initial laws. The pseudorandom stream and its hash establish reproducibility, not independence of ideal observations. The confidence event for the retained R49 family is separate; combining old and new statistical assertions incurs the sum of their failure probabilities.

10. MECHANISM AND ECONOMIC FRONTIERS

Denote witness and FVI by W, F and their final repairs by W^+, F^+ . Pure algebra yields

$$J_W - J_F = (J_W - J_{W^+}) + (J_{W^+} - J_{F^+}) - (J_F - J_{F^+}). \quad (1)$$

All three terms concern actual costs. The first and last are direct measures of removable final-action loss; the middle retains differences from earlier actions, acquired states, continuation approximation and the gate's conservatism. It is not identified as a single causal contribution of neural representation. The experiment reports each term, accepted and blocked cell counts, final action means, final state means and earlier action means by date. This exposes how much of the original ordering can be removed without retraining the continuation. It does not claim that the residual has been exhaustively decomposed into mutually independent mechanisms.

For finite intervals $[L_j, U_j]$, the upper-net-cost choice's true regret is bounded by the selected upper line minus the minimum lower line. To see why this is conservative, consider two overlapping cost intervals: their midpoint ordering may reverse within the same event. The upper-envelope decision remains valid but cannot identify the true minimizer. Pairwise centered intervals can certify individual replacements more sharply than the absolute frontier. For any fixed installation fee, transmission conversion factor and nonnegative resource price, add their declared costs before applying the comparison. A new price does not require new sampling on the same finite event. A new sensing policy does.

11. PROOFS FOR FINITE-SWEEP ACCURACY

PROOF OF THEOREM 7: Fix a pass, date, and state, and suppress their indices. Write J for the incumbent continuation, h for its approximator, π for the incumbent action, and v for the proposal. By the signed evaluation band, for every feasible action a ,

$$\beta a_{t+1} \leq Q(J; x, a) - Q(h; x, a) \leq \beta b_{t+1}.$$

Consequently, the incumbent-relative advantage of the proposal is at most $\bar{d} + \beta w_{t+1}$. An accepted proposal therefore does not raise the incumbent-continuation Bellman cost, while rejection retains exactly that cost. Backward induction gives $J^{\pi^{k+1}} \leq J^{\pi^k}$ at every date. This induction uses the new continuation for the implemented new policy; the comparison itself uses the old continuation.

For an accepted proposal, approximate greediness and the band imply

$$Q(J; x, v) \leq (T_t h)(x) + \alpha + \beta b_{t+1} \leq (T_t J)(x) + \alpha + \beta w_{t+1}.$$

For a rejected proposal, $\bar{d} + \beta w_{t+1} > 0$ and $\bar{d} \leq d + \zeta$. Thus

$$Q(h; x, \pi) < Q(h; x, v) + \zeta + \beta w_{t+1} \leq (T_t h)(x) + \alpha + \zeta + \beta w_{t+1}.$$

Transferring this inequality from h to J adds at most one further βw_{t+1} . Both cases therefore yield

$$T_t^{\pi^{k+1}} J_{t+1}^{\pi^k} \leq T_t J_{t+1}^{\pi^k} + \epsilon_t^k. \quad (2)$$

The already proved policy nonincrease, monotonicity of $T_t^{\pi^{k+1}}$, and (2) give

$$J_t^{\pi^{k+1}} \leq T_t^{\pi^{k+1}} J_{t+1}^{\pi^k} \leq T_t J_{t+1}^{\pi^k} + \epsilon_t^k.$$

The Bellman operator is β_t -Lipschitz in the supremum norm. Subtracting $V_t^* = T_t V_{t+1}^*$ and using feasibility proves (19). Repeated substitution proves (20); at $m = T - t$ its terminal term vanishes. Nonincrease of E_t^k in further passes then gives (21).

The exact case has zero error at every date.

Q.E.D.

PROOF OF PROPOSITION 8: Fix $x \in C$ and any feasible $a \in A_t(x)$. Choose $m \in M_C$ with $\|a - m\| \leq \delta_{A,C}$. Then

$$\begin{aligned} Q_t(h; x, v_C) &\leq Q_t(h; x_C, v_C) + L_t r_C \\ &\leq Q_t(h; x_C, m) + \eta_C + L_t r_C \\ &\leq Q_t(h; x, a) + \eta_C + 2L_t r_C + D_t \delta_{A,C}. \end{aligned}$$

Taking an infimum over a proves the assertion without assuming that the infimum is attained. Menu feasibility is uniform on C by hypothesis.

Q.E.D.

PROOF OF PROPOSITION 9: Take $T = 1$, zero stage cost, $\beta_0 = 1$, and two deterministic actions leading to different terminal states y_0, y_1 . Let $w > 0$ and $0 < \delta < w$. Set

$$h_1(y_0) = w, \quad h_1(y_1) = \delta, \quad g(y_0) = 2w, \quad g(y_1) = \delta.$$

The exact terminal error lies in $[0, w]$. The incumbent chooses y_0 and the exact h_1 -greedy proposal chooses y_1 . Its exact contrast is $d = \delta - w$, with $\alpha = \zeta = 0$. The safe upper comparison is $d + w = \delta > 0$, so the gate retains the incumbent. Its actual optimality gap is $2w - \delta$. Sending δ down to zero establishes sharpness. This example also shows why rejecting an uncertified proposal is not evidence that the proposal lacks a true economic benefit.

Q.E.D.

11.0.0.1. Computed verification and logical scope. The source-bound exact-rational tests use finite transition matrices, signed residual propagation, and independently evaluated policy and optimal costs. They verify policy nonincrease, both gate cases,

the one-pass recurrence, constant-shift invariance, finite-horizon exact termination, and the two-action sharpness construction. They are regression tests for the implementation of the displayed formulas; the preceding arguments, rather than enumeration, prove the theorems for the stated model class. When bands are statistical, all passes and proposed comparisons require a common simultaneous coverage event or a valid sequential error allocation. Reusing a confidence interval outside its stated family does not satisfy the hypotheses.

12. COMPLETE CONSTRUCTION RECORDS

The next table reports all distinct method, economy and resource rungs. Each row has three isolated repetitions with identical policy and certificate checkpoint hashes. Times are the median complete prefix, followed by its minimum and maximum; they do not omit unsuccessful earlier rungs. Counts and storage at full precision are in the machine-readable records. The table's decimal rounding is descriptive: exact rational bounds, not printed rounding, determine attainment.

TABLE I: All distinct construction rungs and complete-prefix times

$d;T;p$	Method	N,K,M	Bound	Median sec.	Range sec.
2;2;1	W	4,2,1	10.0042	0.014	[0.014,0.017]
2;2;1	W	8,2,1	5.591673	0.034	[0.034,0.036]
2;2;1	W	16,4,2	2.795836	0.092	[0.091,0.093]
2;2;1	W	32,4,2	1.692705	0.290	[0.290,0.291]
2;2;1	W	64,8,4	0.8463524	1.458	[1.449,1.459]
2;2;1	W	128,16,8	0.4231762	10.658	[10.612,10.696]
2;2;1	A	4,2,1	10.68755	0.042	[0.042,0.043]
2;2;1	A	8,2,1	6.132349	0.090	[0.090,0.091]
2;2;1	A	16,4,2	3.216146	0.176	[0.176,0.178]
2;2;1	A	32,4,2	1.906506	0.399	[0.398,0.402]
2;2;1	A	64,8,4	0.9622566	1.514	[1.509,1.521]
2;2;1	A	128,16,8	0.4833928	9.817	[9.748,9.900]
2;2;1	F	4,2,1	10.68755	0.029	[0.029,0.029]

	$d; T; p$	Method	N, K, M	Bound	Median sec.	Range sec.	
1							1
2	2;2;1	F	8,2,1	6.132349	0.061	[0.061,0.063]	2
3	2;2;1	F	16,4,2	3.216146	0.119	[0.118,0.122]	3
4	2;2;1	F	32,4,2	1.906506	0.269	[0.268,0.274]	4
5	2;2;1	F	64,8,4	0.9622566	1.134	[1.130,1.143]	5
6	2;2;1	F	128,16,8	0.4833928	8.458	[8.455,8.491]	6
7	2;2;4	W	4,2,1	10.31855	0.014	[0.014,0.014]	7
8	2;2;4	W	8,2,1	5.844473	0.034	[0.034,0.035]	8
9	2;2;4	W	16,4,2	2.922236	0.092	[0.092,0.096]	9
10	2;2;4	W	32,4,2	1.803718	0.290	[0.287,0.303]	10
11	2;2;4	W	64,8,4	0.9018591	1.452	[1.452,1.510]	11
12	2;2;4	W	128,16,8	0.4509295	10.689	[10.584,10.884]	12
13	2;2;4	A	4,2,1	10.92323	0.043	[0.042,0.043]	13
14	2;2;4	A	8,2,1	6.330362	0.091	[0.090,0.092]	14
15	2;2;4	A	16,4,2	3.31506	0.177	[0.176,0.179]	15
16	2;2;4	A	32,4,2	2.000228	0.397	[0.397,0.403]	16
17	2;2;4	A	64,8,4	1.009136	1.509	[1.509,1.524]	17
18	2;2;4	A	128,16,8	0.5068371	9.797	[9.763,9.822]	18
19	2;2;4	F	4,2,1	10.92323	0.029	[0.028,0.033]	19
20	2;2;4	F	8,2,1	6.330362	0.061	[0.060,0.065]	20
21	2;2;4	F	16,4,2	3.31506	0.118	[0.118,0.123]	21
22	2;2;4	F	32,4,2	2.000228	0.268	[0.268,0.273]	22
23	2;2;4	F	64,8,4	1.009136	1.134	[1.131,1.145]	23
24	2;2;4	F	128,16,8	0.5068371	8.453	[8.373,8.520]	24
25	2;3;1	W	4,2,1	14.5143	0.019	[0.019,0.019]	25
26	2;3;1	W	8,2,1	8.228071	0.046	[0.046,0.047]	26
27	2;3;1	W	16,4,2	4.114036	0.126	[0.126,0.129]	27
28	2;3;1	W	32,4,2	2.542479	0.401	[0.398,0.408]	28
29	2;3;1	W	64,8,4	1.271239	2.070	[2.062,2.110]	29
30	2;3;1	W	128,16,8	0.6356197	15.635	[15.607,15.924]	30
31	2;3;1	A	4,2,1	14.97615	0.061	[0.060,0.062]	31
32	2;3;1	A	8,2,1	8.615403	0.130	[0.129,0.131]	32
	2;3;1	A	16,4,2	4.533221	0.252	[0.251,0.255]	

	$d; T; p$	Method	N, K, M	Bound	Median sec.	Range sec.	
1							1
2	2;3;1	A	32,4,2	2.723572	0.568	[0.565,0.574]	2
3	2;3;1	A	64,8,4	1.374878	2.148	[2.145,2.184]	3
4	2;3;1	A	128,16,8	0.6908084	14.176	[14.130,14.444]	4
5	2;3;1	F	4,2,1	14.97615	0.041	[0.041,0.042]	5
6	2;3;1	F	8,2,1	8.615403	0.088	[0.087,0.088]	6
7	2;3;1	F	16,4,2	4.533221	0.171	[0.170,0.192]	7
8	2;3;1	F	32,4,2	2.723572	0.388	[0.388,0.412]	8
9	2;3;1	F	64,8,4	1.374878	1.651	[1.647,1.677]	9
10	2;3;1	F	128,16,8	0.6908084	12.420	[12.409,12.456]	10
11	2;3;4	W	4,2,1	15.02035	0.019	[0.019,0.019]	11
12	2;3;4	W	8,2,1	8.625736	0.046	[0.046,0.048]	12
13	2;3;4	W	16,4,2	4.312868	0.127	[0.125,0.129]	13
14	2;3;4	W	32,4,2	2.714215	0.402	[0.397,0.408]	14
15	2;3;4	W	64,8,4	1.357108	2.085	[2.053,2.103]	15
16	2;3;4	W	128,16,8	0.6785539	15.815	[15.506,15.955]	16
17	2;3;4	A	4,2,1	15.30603	0.061	[0.061,0.061]	17
18	2;3;4	A	8,2,1	8.884068	0.131	[0.130,0.131]	18
19	2;3;4	A	16,4,2	4.667275	0.253	[0.252,0.254]	19
20	2;3;4	A	32,4,2	2.853489	0.569	[0.569,0.569]	20
21	2;3;4	A	64,8,4	1.440006	2.169	[2.165,2.170]	21
22	2;3;4	A	128,16,8	0.723386	14.257	[14.181,14.260]	22
23	2;3;4	F	4,2,1	15.30603	0.042	[0.041,0.042]	23
24	2;3;4	F	8,2,1	8.884068	0.088	[0.087,0.088]	24
25	2;3;4	F	16,4,2	4.667275	0.170	[0.170,0.171]	25
26	2;3;4	F	32,4,2	2.853489	0.390	[0.387,0.391]	26
27	2;3;4	F	64,8,4	1.440006	1.643	[1.637,1.647]	27
28	2;3;4	F	128,16,8	0.723386	12.435	[12.393,12.446]	28
29	3;2;1	W	4,2,1	10.09191	0.033	[0.033,0.033]	29
30	3;2;1	W	8,4,2	5.045957	0.225	[0.224,0.225]	30
31	3;2;1	W	16,8,4	2.522979	2.424	[2.422,2.453]	31
32	3;2;1	W	32,8,4	1.415656	16.615	[16.590,16.644]	32
	3;2;1	F	4,2,1	10.91291	0.096	[0.095,0.096]	

$d; T; p$	Method	N, K, M	Bound	Median sec.	Range sec.
3;2;1	F	8,4,2	5.756259	0.318	[0.318,0.322]
3;2;1	F	16,8,4	2.983083	2.074	[2.059,2.097]
3;2;1	F	32,8,4	1.654898	13.687	[13.539,13.776]
3;3;1	W	4,2,1	14.67402	0.045	[0.044,0.046]
3;3;1	W	8,4,2	7.337012	0.317	[0.313,0.322]
3;3;1	W	16,8,4	3.668506	3.508	[3.476,3.527]
3;3;1	W	32,8,4	2.088657	23.987	[23.952,24.055]
3;3;1	F	4,2,1	15.28673	0.140	[0.140,0.141]
3;3;1	F	8,4,2	8.038961	0.467	[0.463,0.468]
3;3;1	F	16,8,4	4.173215	3.032	[3.027,3.059]
3;3;1	F	32,8,4	2.334703	20.040	[19.984,20.139]
4;2;1	W	4,2,1	10.0042	0.167	[0.164,0.168]
4;2;1	W	8,4,2	5.002099	2.588	[2.561,2.635]
4;2;1	W	16,8,4	2.50105	58.569	[58.203,59.507]
4;2;1	F	4,2,1	10.98489	0.523	[0.520,0.526]
4;2;1	F	8,4,2	5.772966	4.388	[4.386,4.449]
4;2;1	F	16,8,4	2.981482	60.809	[60.637,61.583]
4;3;1	W	4,2,1	14.5143	0.230	[0.230,0.231]
4;3;1	W	8,4,2	7.257149	3.663	[3.649,3.688]
4;3;1	W	16,8,4	3.628575	82.156	[81.795,82.676]
4;3;1	F	4,2,1	15.40485	0.782	[0.776,0.784]
4;3;1	F	8,4,2	8.075833	6.457	[6.444,6.547]
4;3;1	F	16,8,4	4.176837	90.099	[89.833,90.155]

13. ALL DIRECT RETURNED-POLICY COMPARISONS

The labels U, L, C and H denote the uniform initial law and point laws at $(1/8, 1/8)$, $(1/2, 1/2)$ and $(7/8, 7/8)$. Sensor E denotes full information; sensor 8 uses the eight-bit cell contract and downward action spacing. The historical favorable pair is marked R48; other pairs are the actual R49 first crossings. Every interval concerns expected witness cost minus expected FVI cost. All sampled intervals in this table and the next belong to the same new 99 percent family under the independent-bin model.

TABLE II: Every actual first-crossing policy-cost interval

Pair	$T; p$	ε	Law	Bits	Lower	Upper
R49	2;1	4	U	E	0.00543396	0.00798729
R49	2;1	4	U	8	0.00541635	0.00796936
R49	2;1	4	L	E	-0.00156041	0.000833084
R49	2;1	4	L	8	-0.00156513	0.000828396
R49	2;1	4	C	E	0.00372573	0.0061508
R49	2;1	4	C	8	0.00375513	0.00618077
R49	2;1	4	H	E	0.00609378	0.00853627
R49	2;1	4	H	8	0.00589939	0.00833643
R49	2;1	2	U	E	0.0143531	0.0170811
R49	2;1	2	U	8	0.0143519	0.0170783
R49	2;1	2	L	E	0.0547607	0.0574082
R49	2;1	2	L	8	0.0562672	0.0589226
R49	2;1	2	C	E	0.00754534	0.0100938
R49	2;1	2	C	8	0.00764868	0.0101989
R49	2;1	2	H	E	0.00545776	0.00790326
R49	2;1	2	H	8	0.00540822	0.00785334
R49	2;1	1	U	E	0.00608323	0.00862331
R49	2;1	1	U	8	0.00608821	0.00862733
R49	2;1	1	L	E	0.0241465	0.0266462
R49	2;1	1	L	8	0.0243754	0.0268759
R49	2;1	1	C	E	0.0101329	0.0126139
R49	2;1	1	C	8	0.0101188	0.012601
R49	2;1	1	H	E	0.00787923	0.0103208
R49	2;1	1	H	8	0.00773077	0.0101721
R49	2;4	4	U	E	0.0048609	0.00758426
R49	2;4	4	U	8	0.00488276	0.00760722
R49	2;4	4	L	E	0.0185747	0.0211388
R49	2;4	4	L	8	0.0186476	0.0212113
R49	2;4	4	C	E	0.0347841	0.0374079
R49	2;4	4	C	8	0.0346533	0.0372768

	Pair	$T; p$	ε	Law	Bits	Lower	Upper	
1								1
2								2
3	R49	2;4	4	H	E	-8.33958e-05	0.00253601	3
4	R49	2;4	4	H	8	0.000139912	0.00276397	4
5	R49	2;4	2	U	E	0.0126061	0.0154244	5
6	R49	2;4	2	U	8	0.0124777	0.0152934	6
7	R49	2;4	2	L	E	0.0213281	0.0239062	7
8	R49	2;4	2	L	8	0.0216307	0.0242254	8
9	R49	2;4	2	C	E	0.0363241	0.0389543	9
10	R49	2;4	2	C	8	0.0365806	0.0392119	10
11	R49	2;4	2	H	E	0.0408571	0.0435165	11
12	R49	2;4	2	H	8	0.0405602	0.0432247	12
13	R49	2;4	1	U	E	0.00493422	0.0076087	13
14	R49	2;4	1	U	8	0.00498356	0.00765916	14
15	R49	2;4	1	L	E	0.00791584	0.0104887	15
16	R49	2;4	1	L	8	0.00799778	0.010571	16
17	R49	2;4	1	C	E	0.016087	0.018698	17
18	R49	2;4	1	C	8	0.0157071	0.0183196	18
19	R49	2;4	1	H	E	0.00782793	0.0104847	19
20	R49	2;4	1	H	8	0.00787666	0.0105306	20
21	R49	3;1	4	U	E	0.0253656	0.0288147	21
22	R49	3;1	4	U	8	0.0252662	0.0287151	22
23	R49	3;1	4	L	E	0.0788767	0.0824397	23
24	R49	3;1	4	L	8	0.0778317	0.0813824	24
25	R49	3;1	4	C	E	0.100968	0.104201	25
26	R49	3;1	4	C	8	0.101486	0.104716	26
27	R49	3;1	4	H	E	0.0197496	0.0229893	27
28	R49	3;1	4	H	8	0.019754	0.0229886	28
29	R49	3;1	2	U	E	0.0109993	0.0141923	29
30	R49	3;1	2	U	8	0.0110585	0.0142512	30
31	R49	3;1	2	L	E	0.0282946	0.0314077	31
32	R49	3;1	2	L	8	0.0279109	0.0310247	32
	R49	3;1	2	C	E	0.0176355	0.0207641	
	R49	3;1	2	C	8	0.0171906	0.0203209	

	Pair	$T;p$	ε	Law	Bits	Lower	Upper	
1								1
2								2
3	R49	3;1	2	H	E	0.0126606	0.0156975	3
4	R49	3;1	2	H	8	0.0126999	0.0157435	4
5	R49	3;1	1	U	E	0.00423544	0.00731756	5
6	R49	3;1	1	U	8	0.00451493	0.0075994	6
7	R49	3;1	1	L	E	0.00514129	0.008227	7
8	R49	3;1	1	L	8	0.00545945	0.00854446	8
9	R49	3;1	1	C	E	0.00460682	0.00763251	9
10	R49	3;1	1	C	8	0.00492024	0.00794768	10
11	R49	3;1	1	H	E	0.00035797	0.00337421	11
12	R49	3;1	1	H	8	0.00036318	0.00337929	12
13	R49	3;4	4	U	E	0.0186753	0.022252	13
14	R49	3;4	4	U	8	0.0186419	0.0222187	14
15	R49	3;4	4	L	E	0.0276014	0.0309773	15
16	R49	3;4	4	L	8	0.0279277	0.0313061	16
17	R49	3;4	4	C	E	0.0549145	0.058394	17
18	R49	3;4	4	C	8	0.0535423	0.0569789	18
19	R49	3;4	4	H	E	0.0500905	0.0536733	19
20	R49	3;4	4	H	8	0.0501068	0.0536738	20
21	R49	3;4	2	U	E	0.00722022	0.0105987	21
22	R49	3;4	2	U	8	0.00727216	0.0106525	22
23	R49	3;4	2	L	E	0.0111161	0.0144007	23
24	R49	3;4	2	L	8	0.0111398	0.0144261	24
25	R49	3;4	2	C	E	0.0119992	0.0152703	25
26	R49	3;4	2	C	8	0.0120479	0.0153198	26
27	R49	3;4	2	H	E	0.00896933	0.0122714	27
28	R49	3;4	2	H	8	0.00896107	0.0122625	28
29	R49	3;4	1	U	E	0.00231571	0.00560954	29
30	R49	3;4	1	U	8	0.00249689	0.00579256	30
31	R49	3;4	1	L	E	0.00405598	0.00731121	31
32	R49	3;4	1	L	8	0.000239139	0.00350007	32
	R49	3;4	1	C	E	0.00285917	0.00611809	
	R49	3;4	1	C	8	0.00270615	0.00596212	

Pair	$T;p$	ε	Law	Bits	Lower	Upper
R49	3;4	1	H	E	0.000774402	0.00403881
R49	3;4	1	H	8	0.000787911	0.0040506
R48	3;4	2	U	E	-0.00139342	0.00183546
R48	3;4	2	U	8	-0.00138819	0.00184069
R48	3;4	2	L	E	-0.00148104	0.00174625
R48	3;4	2	L	8	-0.00148102	0.00174626
R48	3;4	2	C	E	-0.00135341	0.0018749
R48	3;4	2	C	8	-0.00134267	0.00188569
R48	3;4	2	H	E	-0.00139997	0.00182783
R48	3;4	2	H	8	-0.00141355	0.00181414

Endpoints are displayed to six significant digits; exact rational endpoints and policy hashes govern every sign and fee conclusion. R48 rows evaluate the old favorable N=32 witness versus N=64 FVI pair under fresh independent interval paths.

14. ALL SENSOR-POLICY CONTRASTS

Each row compares the actual b -bit implementation with the same returned controller at 16 bits, under the uniform initial law. The 16-bit self-rows are identities. The corresponding price selectors and exact continuous-price envelopes use these rows without additional sampling. Upper-net-cost regret is relative to the best bit in this catalogue, not an unobserved optimum over continuous precision or arbitrary sensors.

TABLE III: Every actual sensor-cost difference relative to 16 bits

$T;p$	Method	Bits	Lower	Upper
2;1	W	4	0.000261375	0.00285711
2;1	W	5	-0.000427494	0.00215671
2;1	W	6	-0.00204196	0.000521085
2;1	W	7	-0.00088098	0.00160987
2;1	W	8	-0.00122806	0.00124059

	$T;p$	Method	Bits	Lower	Upper	
1						1
2						2
3	2;1	W	9	-0.00121323	0.00123098	3
4	2;1	W	10	-0.00120642	0.00122091	4
5	2;1	W	11	-0.00119893	0.00121541	5
6	2;1	W	12	-0.00120501	0.00120022	6
7	2;1	W	13	-0.0011989	0.00119995	7
8	2;1	W	14	-0.00119712	0.00119717	8
9	2;1	W	15	-0.00119517	0.00119587	9
10	2;1	W	16	0	0	10
11	2;1	F	4	0.000199127	0.0026252	11
12	2;1	F	5	-0.000584413	0.00181963	12
13	2;1	F	6	-0.00085593	0.00153971	13
14	2;1	F	7	-0.00111845	0.00126841	14
15	2;1	F	8	-0.00117522	0.00120999	15
16	2;1	F	9	-0.00118398	0.00120093	16
17	2;1	F	10	-0.00118806	0.00119669	17
18	2;1	F	11	-0.00119023	0.00119442	18
19	2;1	F	12	-0.00119128	0.00119331	19
20	2;1	F	13	-0.00119182	0.00119274	20
21	2;1	F	14	-0.00119207	0.00119247	21
22	2;1	F	15	-0.00119219	0.00119235	22
23	2;1	F	16	0	0	23
24	2;4	W	4	-4.77011e-05	0.00267196	24
25	2;4	W	5	-0.000905067	0.00180125	25
26	2;4	W	6	-0.00220586	0.000485462	26
27	2;4	W	7	-0.00108854	0.00153925	27
28	2;4	W	8	-0.00124312	0.00137376	28
29	2;4	W	9	-0.00131086	0.00128822	29
30	2;4	W	10	-0.00128939	0.00129611	30
31	2;4	W	11	-0.00128416	0.00129219	31
32	2;4	W	12	-0.00128539	0.00128378	32
	2;4	W	13	-0.00128173	0.0012827	
	2;4	W	14	-0.00128018	0.00128022	

	$T; p$	Method	Bits	Lower	Upper	
1						1
2						2
3	2;4	W	15	-0.00127809	0.00128043	3
4	2;4	W	16	0	0	4
5	2;4	F	4	-0.00112704	0.00143668	5
6	2;4	F	5	-0.00121844	0.00133951	6
7	2;4	F	6	-0.00125571	0.00129926	7
8	2;4	F	7	-0.00126208	0.00129244	8
9	2;4	F	8	-0.00127554	0.00127722	9
10	2;4	F	9	-0.00127593	0.00127681	10
11	2;4	F	10	-0.00127615	0.00127658	11
12	2;4	F	11	-0.00127626	0.00127646	12
13	2;4	F	12	-0.0012763	0.00127642	13
14	2;4	F	13	-0.00127634	0.00127638	14
15	2;4	F	14	-0.00127635	0.00127637	15
16	2;4	F	15	-0.00127636	0.00127636	16
17	2;4	F	16	0	0	17
18	3;1	W	4	-0.000101036	0.00300605	18
19	3;1	W	5	-0.00041666	0.00269816	19
20	3;1	W	6	-0.000850866	0.00225551	20
21	3;1	W	7	-0.0023459	0.000745795	21
22	3;1	W	8	-0.00124523	0.00180352	22
23	3;1	W	9	-0.00147051	0.00156613	23
24	3;1	W	10	-0.00150886	0.00151075	24
25	3;1	W	11	-0.00150001	0.00150822	25
26	3;1	W	12	-0.00150165	0.00149869	26
27	3;1	W	13	-0.00149858	0.00149581	27
28	3;1	W	14	-0.00149663	0.00149349	28
29	3;1	W	15	-0.00149339	0.00149403	29
30	3;1	W	16	0	0	30
31	3;1	F	4	0.000691523	0.00372718	31
32	3;1	F	5	-0.000505628	0.00250065	32
	3;1	F	6	-0.0010836	0.00190806	
	3;1	F	7	-0.00122457	0.00176312	

	$T;p$	Method	Bits	Lower	Upper	
1	3;1	F	8	-0.001473	0.00150813	1
2	3;1	F	9	-0.0014817	0.00149918	2
3	3;1	F	10	-0.0014861	0.00149463	3
4	3;1	F	11	-0.00148819	0.00149247	4
5	3;1	F	12	-0.00148929	0.00149132	5
6	3;1	F	13	-0.00148981	0.00149079	6
7	3;1	F	14	-0.00149008	0.0014905	7
8	3;1	F	15	-0.00149022	0.00149036	8
9	3;1	F	16	0	0	9
10	3;4	W	4	-0.00115463	0.00216779	10
11	3;4	W	5	-0.00128854	0.00202944	11
12	3;4	W	6	-0.0012672	0.00204406	12
13	3;4	W	7	-0.00203778	0.00126435	13
14	3;4	W	8	-0.00147323	0.00179482	14
15	3;4	W	9	-0.00162558	0.00163607	15
16	3;4	W	10	-0.00162825	0.00162325	16
17	3;4	W	11	-0.00162181	0.00162209	17
18	3;4	W	12	-0.00161759	0.00162064	18
19	3;4	W	13	-0.00161723	0.00161692	19
20	3;4	W	14	-0.00161637	0.00161497	20
21	3;4	W	15	-0.00161356	0.00161547	21
22	3;4	W	16	0	0	22
23	3;4	F	4	-0.00143754	0.00179918	23
24	3;4	F	5	-0.0015455	0.00168516	24
25	3;4	F	6	-0.0015881	0.00163939	25
26	3;4	F	7	-0.00159646	0.0016305	26
27	3;4	F	8	-0.00161162	0.00161347	27
28	3;4	F	9	-0.00161206	0.00161302	28
29	3;4	F	10	-0.00161229	0.00161278	29
30	3;4	F	11	-0.00161241	0.00161265	30
31	3;4	F	12	-0.00161247	0.00161258	31
32	3;4	F	13	-0.0016125	0.00161255	32

$T;p$	Method	Bits	Lower	Upper
3;4	F	14	-0.00161251	0.00161254
3;4	F	15	-0.00161252	0.00161253
3;4	F	16	0	0

All rows use the uniform initial law and the actual first-target-one controller. Self-comparisons are exact identities. Both policies use the declared robust repair and action quantum.

15. RECONSTRUCTION AND PRESERVATION

The release verifies frozen science-source identities, every recorded checkpoint, all signed policy formulas, every first crossing, complete-prefix counts, repeated policy identities and the confidence intervals reconstructed from endpoint moments. Additional exact tests check the primitive regions, resource allocation, original-owner identity, fee semantics and net-choice bound. The complete development article and supplement retain all baseline labeled results; the original paths and adverse results are unchanged. A clean Git-archive rebuild removes generated PDFs and derived tables, then rebuilds from ordinary committed sources and frozen records without network or new scientific execution. Compilation, finite tests and source identities do not constitute independent mathematical peer approval.

16. EVERY R50 ACTUAL-COST INTERVAL AND MECHANISM RECORD

The following tables use the corrected outward-subtraction replay. C denotes the separate common-accuracy block, P the primary target-five plan, Q the primary target-two plan, O the original R49 target-two policies, A residual-driven FVI, and I the isotropic block. U is the uniform initial law; L, M and H denote point laws at $1/8, 1/2$ and $7/8$ in every coordinate. Each group has four absolute costs and six contrasts; the table lists all 280 estimands. Printed endpoints are rounded for display. Exact rational endpoints and source-bound moments govern inference.

TABLE IV: Every new absolute expected policy-cost interval

Case	Policy	Lower	Upper
C2;2;1;U	left	0.6242	0.6378
C2;2;1;U	left-repaired	0.61117	0.62482
C2;2;1;U	right	0.59821	0.61177
C2;2;1;U	right-repaired	0.59628	0.60988
C3;2;1;U	left	0.58992	0.60068
C3;2;1;U	left-repaired	0.58062	0.59146
C3;2;1;U	right	0.56904	0.57982
C3;2;1;U	right-repaired	0.56768	0.57849
C4;2;1;U	left	0.62188	0.63199
C4;2;1;U	left-repaired	0.60399	0.61391
C4;2;1;U	right	0.5855	0.59525
C4;2;1;U	right-repaired	0.5854	0.59515
I2;2;1;U	left	0.63959	0.65386
I2;2;1;U	left-repaired	0.61456	0.6285
I2;2;1;U	right	0.59407	0.60781
I2;2;1;U	right-repaired	0.59363	0.60737
O2;2;1;M	left	0.30128	0.30382
O2;2;1;M	left-repaired	0.29236	0.29488
O2;2;1;M	right	0.29238	0.29489
O2;2;1;M	right-repaired	0.29236	0.29488
O2;2;1;L	left	1.5737	1.5766
O2;2;1;L	left-repaired	1.5626	1.565
O2;2;1;L	right	1.5163	1.5188
O2;2;1;L	right-repaired	1.5162	1.5186
O2;2;1;H	left	0.19157	0.19409
O2;2;1;H	left-repaired	0.18888	0.19135
O2;2;1;H	right	0.18497	0.18741
O2;2;1;H	right-repaired	0.18423	0.18668
O2;2;1;U	left	0.61089	0.62464
O2;2;1;U	left-repaired	0.60392	0.61765

	Case	Policy	Lower	Upper	
1					1
2					2
3	O2;2;1;U	right	0.59513	0.60882	3
4	O2;2;1;U	right-repaired	0.59491	0.6086	4
5	O2;2;4;M	left	0.42746	0.43017	5
6	O2;2;4;M	left-repaired	0.42249	0.42507	6
7	O2;2;4;M	right	0.38962	0.39224	7
8	O2;2;4;M	right-repaired	0.38952	0.39214	8
9	O2;2;4;L	left	1.6588	1.6614	9
10	O2;2;4;L	left-repaired	1.6564	1.659	10
11	O2;2;4;L	right	1.6358	1.6385	11
12	O2;2;4;L	right-repaired	1.6358	1.6385	12
13	O2;2;4;H	left	0.2474	0.2501	13
14	O2;2;4;H	left-repaired	0.24264	0.24526	14
15	O2;2;4;H	right	0.20559	0.20818	15
16	O2;2;4;H	right-repaired	0.20554	0.20813	16
17	O2;2;4;U	left	0.69261	0.7071	17
18	O2;2;4;U	left-repaired	0.68704	0.70155	18
19	O2;2;4;U	right	0.67867	0.69319	19
20	O2;2;4;U	right-repaired	0.67849	0.69301	20
21	O2;3;1;M	left	0.42649	0.42982	21
22	O2;3;1;M	left-repaired	0.42326	0.42657	22
23	O2;3;1;M	right	0.40778	0.41096	23
24	O2;3;1;M	right-repaired	0.40776	0.41095	24
25	O2;3;1;L	left	1.6968	1.7	25
26	O2;3;1;L	left-repaired	1.6932	1.6964	26
27	O2;3;1;L	right	1.6673	1.6705	27
28	O2;3;1;L	right-repaired	1.6672	1.6704	28
29	O2;3;1;H	left	0.29136	0.29449	29
30	O2;3;1;H	left-repaired	0.28954	0.29265	30
31	O2;3;1;H	right	0.27712	0.28025	31
32	O2;3;1;H	right-repaired	0.27696	0.28009	32
	O2;3;1;U	left	0.71942	0.73411	
	O2;3;1;U	left-repaired	0.71638	0.73106	

1	Case	Policy	Lower	Upper	1
2					2
3	O2;3;1;U	right	0.70681	0.72143	3
4	O2;3;1;U	right-repaired	0.70676	0.72139	4
5	O2;3;4;M	left	0.58099	0.58436	5
6	O2;3;4;M	left-repaired	0.57925	0.58262	6
7	O2;3;4;M	right	0.56732	0.57065	7
8	O2;3;4;M	right-repaired	0.56709	0.57042	8
9	O2;3;4;L	left	1.8582	1.8616	9
10	O2;3;4;L	left-repaired	1.856	1.8594	10
11	O2;3;4;L	right	1.8454	1.8487	11
12	O2;3;4;L	right-repaired	1.8454	1.8487	12
13	O2;3;4;H	left	0.35451	0.35788	13
14	O2;3;4;H	left-repaired	0.35176	0.35509	14
15	O2;3;4;H	right	0.34392	0.34723	15
16	O2;3;4;H	right-repaired	0.34391	0.34721	16
17	O2;3;4;U	left	0.86235	0.87799	17
18	O2;3;4;U	left-repaired	0.86048	0.87613	18
19	O2;3;4;U	right	0.85339	0.86906	19
20	O2;3;4;U	right-repaired	0.85323	0.86891	20
21	Q2;2;1;U	left	0.60972	0.62332	21
22	Q2;2;1;U	left-repaired	0.60276	0.61638	22
23	Q2;2;1;U	right	0.59518	0.60882	23
24	Q2;2;1;U	right-repaired	0.59444	0.6081	24
25	P2;2;1;U	left	0.6222	0.63577	25
26	P2;2;1;U	left-repaired	0.60912	0.62274	26
27	P2;2;1;U	right	0.59639	0.60993	27
28	P2;2;1;U	right-repaired	0.59445	0.60803	28
29	Q2;2;4;U	left	0.69243	0.70702	29
30	Q2;2;4;U	left-repaired	0.68754	0.70213	30
31	Q2;2;4;U	right	0.68106	0.69562	31
32	Q2;2;4;U	right-repaired	0.68086	0.69542	32
	Q2;3;1;U	left	0.72237	0.73696	
	Q2;3;1;U	left-repaired	0.71906	0.73365	

Case	Policy	Lower	Upper
Q2;3;1;U	right	0.70782	0.7224
Q2;3;1;U	right-repaired	0.70769	0.72228
Q2;3;4;U	left	0.86908	0.88468
Q2;3;4;U	left-repaired	0.86539	0.881
Q2;3;4;U	right	0.85511	0.87074
Q2;3;4;U	right-repaired	0.85451	0.87015
P3;2;1;U	left	0.58981	0.60055
P3;2;1;U	left-repaired	0.58046	0.59128
P3;2;1;U	right	0.56902	0.57978
P3;2;1;U	right-repaired	0.56764	0.57844
P4;2;1;U	left	0.61951	0.62957
P4;2;1;U	left-repaired	0.60168	0.61155
P4;2;1;U	right	0.58357	0.59328
P4;2;1;U	right-repaired	0.58332	0.59303
A2;2;1;U	left	0.59772	0.61129
A2;2;1;U	left-repaired	0.59585	0.60945
A2;2;1;U	right	0.59777	0.61134
A2;2;1;U	right-repaired	0.59585	0.60946

TABLE V: Every new direct expected policy-cost contrast

Case	Contrast	Lower	Upper
C2;2;1;U	L minus L+	0.011448	0.014567
C2;2;1;U	L minus R	0.023033	0.028988
C2;2;1;U	L minus R+	0.024948	0.030894
C2;2;1;U	L+ minus R	0.010133	0.015872
C2;2;1;U	L+ minus R+	0.012051	0.017775
C2;2;1;U	R minus R+	0.00067113	0.0031495
C3;2;1;U	L minus L+	0.0077535	0.010757
C3;2;1;U	L minus R	0.01796	0.023776

1	Case	Contrast	Lower	Upper	1
2					2
3	C3;2;1;U	L minus R+	0.019309	0.025125	3
4	C3;2;1;U	L+ minus R	0.0087758	0.014449	4
5	C3;2;1;U	L+ minus R+	0.010129	0.015794	5
6	C3;2;1;U	R minus R+	0.0001224	0.0025755	6
7	C4;2;1;U	L minus L+	0.016432	0.019542	7
8	C4;2;1;U	L minus R	0.033636	0.039473	8
9	C4;2;1;U	L minus R+	0.033741	0.039578	9
10	C4;2;1;U	L+ minus R	0.015807	0.021327	10
11	C4;2;1;U	L+ minus R+	0.015912	0.021433	11
12	C4;2;1;U	R minus R+	0	0.0012997	12
13	I2;2;1;U	L minus L+	0.023541	0.026849	13
14	I2;2;1;U	L minus R	0.042713	0.048868	14
15	I2;2;1;U	L minus R+	0.043151	0.049305	15
16	I2;2;1;U	L+ minus R	0.017772	0.023418	16
17	I2;2;1;U	L+ minus R+	0.018209	0.023856	17
18	I2;2;1;U	R minus R+	0	0.00164	18
19	O2;2;1;M	L minus L+	0.007622	0.010239	19
20	O2;2;1;M	L minus R	0.0064146	0.011415	20
21	O2;2;1;M	L minus R+	0.00643	0.011431	21
22	O2;2;1;M	L+ minus R	-0.0023997	0.002369	22
23	O2;2;1;M	L+ minus R+	-0.0023843	0.0023843	23
24	O2;2;1;M	R minus R+	0	0.0012077	24
25	O2;2;1;L	L minus L+	0.0099411	0.012722	25
26	O2;2;1;L	L minus R	0.054972	0.060123	26
27	O2;2;1;L	L minus R+	0.055155	0.060307	27
28	O2;2;1;L	L+ minus R	0.043822	0.048609	28
29	O2;2;1;L	L+ minus R+	0.044007	0.048792	29
30	O2;2;1;L	R minus R+	0	0.001381	30
31	O2;2;1;H	L minus L+	0.0014934	0.003946	31
32	O2;2;1;H	L minus R	0.0042173	0.0090667	32
	O2;2;1;H	L minus R+	0.0049502	0.0097948	
	O2;2;1;H	L+ minus R	0.001523	0.0063216	

	Case	Contrast	Lower	Upper	
1					1
2					2
3	O2;2;1;H	L+ minus R+	0.0022567	0.007049	3
4	O2;2;1;H	R minus R+	0	0.0019272	4
5	O2;2;1;U	L minus L+	0.0056474	0.0083165	5
6	O2;2;1;U	L minus R	0.013161	0.018416	6
7	O2;2;1;U	L minus R+	0.013385	0.01864	7
8	O2;2;1;U	L+ minus R	0.0062318	0.011381	8
9	O2;2;1;U	L+ minus R+	0.0064562	0.011605	9
10	O2;2;1;U	R minus R+	0	0.0014206	10
11	O2;2;4;M	L minus L+	0.0036933	0.0063836	11
12	O2;2;4;M	L minus R	0.035287	0.040496	12
13	O2;2;4;M	L minus R+	0.035387	0.040597	13
14	O2;2;4;M	L+ minus R	0.03028	0.035426	14
15	O2;2;4;M	L+ minus R+	0.030381	0.035526	15
16	O2;2;4;M	R minus R+	0	0.0013773	16
17	O2;2;4;L	L minus L+	0.0010828	0.0036974	17
18	O2;2;4;L	L minus R	0.020345	0.025506	18
19	O2;2;4;L	L minus R+	0.020345	0.025506	19
20	O2;2;4;L	L+ minus R	0.017976	0.023095	20
21	O2;2;4;L	L+ minus R+	0.017976	0.023095	21
22	O2;2;4;L	R minus R+	0	0.0012764	22
23	O2;2;4;H	L minus L+	0.0034416	0.006155	23
24	O2;2;4;H	L minus R	0.039236	0.044497	24
25	O2;2;4;H	L minus R+	0.039288	0.044549	25
26	O2;2;4;H	L+ minus R	0.034497	0.039639	26
27	O2;2;4;H	L+ minus R+	0.034549	0.039692	27
28	O2;2;4;H	R minus R+	0	0.0013293	28
29	O2;2;4;U	L minus L+	0.004188	0.0069254	29
30	O2;2;4;U	L minus R	0.011181	0.016658	30
31	O2;2;4;U	L minus R+	0.011361	0.016838	31
32	O2;2;4;U	L+ minus R	0.0056495	0.011076	32
	O2;2;4;U	L+ minus R+	0.0058295	0.011256	
	O2;2;4;U	R minus R+	0	0.0014585	

1	Case	Contrast	Lower	Upper	1
2					2
3	O2;3;1;M	L minus L+	0.0016956	0.0047777	3
4	O2;3;1;M	L minus R	0.015697	0.021868	4
5	O2;3;1;M	L minus R+	0.015714	0.021886	5
6	O2;3;1;M	L+ minus R	0.012467	0.018624	6
7	O2;3;1;M	L+ minus R+	0.012485	0.018641	7
8	O2;3;1;M	R minus R+	0	0.0015083	8
9	O2;3;1;L	L minus L+	0.0020031	0.005116	9
10	O2;3;1;L	L minus R	0.026367	0.032516	10
11	O2;3;1;L	L minus R+	0.026477	0.032626	11
12	O2;3;1;L	L+ minus R	0.022819	0.028945	12
13	O2;3;1;L	L+ minus R+	0.022928	0.029055	13
14	O2;3;1;L	R minus R+	0	0.0016012	14
15	O2;3;1;H	L minus L+	0.00030976	0.0033545	15
16	O2;3;1;H	L minus R	0.011221	0.017268	16
17	O2;3;1;H	L minus R+	0.011378	0.017424	17
18	O2;3;1;H	L+ minus R	0.0093942	0.01543	18
19	O2;3;1;H	L+ minus R+	0.0095515	0.015586	19
20	O2;3;1;H	R minus R+	0	0.0016481	20
21	O2;3;1;U	L minus L+	0.0014967	0.0045926	21
22	O2;3;1;U	L minus R	0.0095127	0.015773	22
23	O2;3;1;U	L minus R+	0.0095603	0.01582	23
24	O2;3;1;U	L+ minus R	0.0064826	0.012713	24
25	O2;3;1;U	L+ minus R+	0.0065302	0.012761	25
26	O2;3;1;U	R minus R+	0	0.0015385	26
27	O2;3;4;M	L minus L+	9.4671e-05	0.0033779	27
28	O2;3;4;M	L minus R	0.010433	0.016948	28
29	O2;3;4;M	L minus R+	0.01066	0.017175	29
30	O2;3;4;M	L+ minus R	0.0087064	0.015202	30
31	O2;3;4;M	L+ minus R+	0.0089337	0.015429	31
32	O2;3;4;M	R minus R+	0	0.00184	32
	O2;3;4;L	L minus L+	0.0005048	0.0037993	
	O2;3;4;L	L minus R	0.0095446	0.01608	

	Case	Contrast	Lower	Upper	
1					1
2					2
3	O2;3;4;L	L minus R+	0.0095504	0.016086	3
4	O2;3;4;L	L+ minus R	0.0074107	0.01391	4
5	O2;3;4;L	L+ minus R+	0.0074164	0.013916	5
6	O2;3;4;L	R minus R+	0	0.0016183	6
7	O2;3;4;H	L minus L+	0.0011154	0.0044219	7
8	O2;3;4;H	L minus R	0.0073415	0.013898	8
9	O2;3;4;H	L minus R+	0.0073538	0.013911	9
10	O2;3;4;H	L+ minus R	0.004596	0.011107	10
11	O2;3;4;H	L+ minus R+	0.0046083	0.011119	11
12	O2;3;4;H	R minus R+	0	0.0016249	12
13	O2;3;4;U	L minus L+	0.00021625	0.0035063	13
14	O2;3;4;U	L minus R	0.0056055	0.012273	14
15	O2;3;4;U	L minus R+	0.0057651	0.012433	15
16	O2;3;4;U	L+ minus R	0.0037509	0.010406	16
17	O2;3;4;U	L+ minus R+	0.0039105	0.010565	17
18	O2;3;4;U	R minus R+	0	0.001774	18
19	Q2;2;1;U	L minus L+	0.0055893	0.0083122	19
20	Q2;2;1;U	L minus R	0.01186	0.017187	20
21	Q2;2;1;U	L minus R+	0.01259	0.017917	21
22	Q2;2;1;U	L+ minus R	0.0049684	0.010177	22
23	Q2;2;1;U	L+ minus R+	0.0056983	0.010907	23
24	Q2;2;1;U	R minus R+	0	0.0019382	24
25	P2;2;1;U	L minus L+	0.011497	0.014615	25
26	P2;2;1;U	L minus R	0.022852	0.028801	26
27	P2;2;1;U	L minus R+	0.024778	0.030719	27
28	P2;2;1;U	L+ minus R	0.0099052	0.015635	28
29	P2;2;1;U	L+ minus R+	0.011835	0.01755	29
30	P2;2;1;U	R minus R+	0.00068331	0.0031614	30
31	Q2;2;4;U	L minus L+	0.003535	0.0062382	31
32	Q2;2;4;U	L minus R	0.0086864	0.014074	32
	Q2;2;4;U	L minus R+	0.0088869	0.014274	
	Q2;2;4;U	L+ minus R	0.0038216	0.0091655	

	Case	Contrast	Lower	Upper	
1					1
2					2
3	Q2;2;4;U	L+ minus R+	0.004022	0.0093656	3
4	Q2;2;4;U	R minus R+	0	0.0014794	4
5	Q2;3;1;U	L minus L+	0.0017403	0.0048791	5
6	Q2;3;1;U	L minus R	0.01136	0.017753	6
7	Q2;3;1;U	L minus R+	0.011488	0.01788	7
8	Q2;3;1;U	L+ minus R	0.0080678	0.014426	8
9	Q2;3;1;U	L+ minus R+	0.0081953	0.014553	9
10	Q2;3;1;U	R minus R+	0	0.0016207	10
11	Q2;3;4;U	L minus L+	0.0020091	0.0053568	11
12	Q2;3;4;U	L minus R	0.010559	0.017349	12
13	Q2;3;4;U	L minus R+	0.011154	0.017943	13
14	Q2;3;4;U	L+ minus R	0.006888	0.013654	14
15	Q2;3;4;U	L+ minus R+	0.0074835	0.014248	15
16	Q2;3;4;U	R minus R+	0	0.0022148	16
17	P3;2;1;U	L minus L+	0.0078066	0.010814	17
18	P3;2;1;U	L minus R	0.017874	0.023687	18
19	P3;2;1;U	L minus R+	0.019232	0.025044	19
20	P3;2;1;U	L+ minus R	0.008637	0.014304	20
21	P3;2;1;U	L+ minus R+	0.0099988	0.015657	21
22	P3;2;1;U	R minus R+	0.00013073	0.0025845	22
23	P4;2;1;U	L minus L+	0.016374	0.019482	23
24	P4;2;1;U	L minus R	0.033196	0.039029	24
25	P4;2;1;U	L minus R+	0.033446	0.039279	25
26	P4;2;1;U	L+ minus R	0.015426	0.020943	26
27	P4;2;1;U	L+ minus R+	0.015676	0.021193	27
28	P4;2;1;U	R minus R+	0	0.0014477	28
29	A2;2;1;U	L minus L+	0.00062241	0.0030972	29
30	A2;2;1;U	L minus R	-0.0024517	0.0023493	30
31	A2;2;1;U	L minus R+	-0.00057956	0.0042851	31
32	A2;2;1;U	L+ minus R	-0.0043435	0.00052149	32
	A2;2;1;U	L+ minus R+	-0.0024125	0.0023985	
	A2;2;1;U	R minus R+	0.00066516	0.0031428	

Both tables are reconstructed from stored outward means, variance bounds and deterministic supports. Numerical replay uses the same original samples. Statistical interpretation requires the independent-bin model.

TABLE VI: Final-action mechanism diagnostics

Case	Policy	Accepted	Blocked	Old action	New action	Derivative calls
C2;2;1;U	L	81251	1686	0.13805	0.16199	1216761
C2;2;1;U	R	92005	2261	0.17813	0.16288	1223163
C3;2;1;U	L	71835	1430	0.15264	0.16793	1212973
C3;2;1;U	R	86739	1858	0.18142	0.16825	1220030
C4;2;1;U	L	105827	1124	0.11931	0.16893	1209765
C4;2;1;U	R	79985	9020	0.16899	0.16935	1219031
I2;2;1;U	L	115800	1824	0.10471	0.16094	1216727
I2;2;1;U	R	87457	6803	0.16155	0.16239	1222172
O2;2;1;M	L	123097	0	0.13434	0.18236	1179648
O2;2;1;M	R	66240	0	0.18188	0.18236	1179648
O2;2;1;L	L	51516	0	0.13357	0.14812	1179648
O2;2;1;L	R	25162	0	0.14947	0.14977	1179648
O2;2;1;H	L	131072	0	0.11338	0.11033	1301723
O2;2;1;H	R	131072	0	0.10626	0.12107	1286030
O2;2;1;U	L	96199	2180	0.13976	0.16238	1217297
O2;2;1;U	R	86179	5606	0.16295	0.16247	1223021
O2;2;4;M	L	131072	0	0.06499	0.09143	1253562
O2;2;4;M	R	131072	0	0.08886	0.08423	1245289
O2;2;4;L	L	131072	0	0.12539	0.13479	1227082
O2;2;4;L	R	0	131072	0.13106	0.13106	1232486
O2;2;4;H	L	131072	0	0.04179	0.04061	1294458
O2;2;4;H	R	131072	0	0.05060	0.04725	1278365
O2;2;4;U	L	127091	2453	0.08431	0.08510	1255705
O2;2;4;U	R	116259	12245	0.08476	0.08500	1255578
O2;3;1;M	L	74044	0	0.15964	0.17640	1179648
O2;3;1;M	R	50391	0	0.17885	0.17907	1179648
O2;3;1;L	L	47744	0	0.14872	0.15570	1179648

Case	Policy	Accepted	Blocked	Old action	New action	Derivative calls
1						
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Means are descriptive over the recorded paths, not confidence intervals for separate mechanism parameters. Complete date-wise earlier actions, terminal state vectors, capacity and quantum repair counts, and exact fallbacks are deposited.

TABLE VII: New-construction operation and storage account

Case	Method	Target eval.	Actor scalars	KiB	Coeff. bits	MiB
C2;2;1;U	W	1,156	578	41.8	36	41.4
C2;2;1;U	F	1,156	578	26.1	42	41.7
C3;2;1;U	W	19,652	9,826	851.1	42	44.3
C3;2;1;U	F	19,652	9,826	434.4	42	42.3
C4;2;1;U	W	236,196	13,122	1024.5	42	50.5
C4;2;1;U	F	3,242,952	167,042	7464.3	42	169.8
I2;2;1;U	W	11,664	162	14.2	41	42.4
I2;2;1;U	F	11,664	162	9.3	41	42.0
Q2;2;1;U	W	13,068	2,178	171.9	41	41.6
Q2;2;1;U	F	13,068	2,178	98.6	42	42.1
P2;2;1;U	W	1,156	578	41.8	36	42.0
P2;2;1;U	F	1,156	578	26.1	42	42.0
Q2;2;4;U	W	19,602	2,178	178.4	42	41.6
Q2;2;4;U	F	19,602	2,178	99.4	42	42.0
Q2;3;1;U	W	126,750	12,675	1005.0	42	45.8
Q2;3;1;U	F	126,750	12,675	550.0	42	41.9
Q2;3;4;U	W	126,750	12,675	983.8	42	45.8
Q2;3;4;U	F	126,750	12,675	539.4	42	41.8
P3;2;1;U	W	19,652	9,826	851.1	42	44.5
P3;2;1;U	F	19,652	9,826	434.4	42	42.5
P4;2;1;U	W	236,196	13,122	1024.5	42	51.3
P4;2;1;U	F	236,196	13,122	575.2	42	48.3
A2;2;1;U	A	4,472	578	39.5	42	41.9
A2;2;1;U	F	1,156	578	26.1	42	42.0

Target evaluations include every failed common-accuracy attempt. Actor scalars and checkpoint KiB describe the returned model. Coefficient bits are the true maximum, not the additive descriptor in the

raw prefix map; MiB is process peak memory. Full operation categories, including terminal gates and exact fallbacks, remain in JSON.

17. NUMERICAL CORRECTION AND SEQUENTIAL COMMON-ACCURACY DESIGN

The primary shared allocation is sufficient for the witness bound, not automatically for conventional FVI's actual interpolated modulus. Its failed certificates are retained. The common-accuracy amendment was specified after that outcome and frozen before its own execution; it charges every failed construction before doubling the state quota. It is not a retrospective recoding of the primary study.

During publication review, nearest-rounded subtraction of different policies' already outward path endpoints was replaced by directed outward subtraction. The same bin streams, policies and sample counts were replayed. Original sources and results remain unchanged; a separate correction manifest binds the new evaluator, and the audit checks identical streams and unchanged sign classifications. Replaying the same observations adds no statistical sample size. The additive raw prefix field for coefficient-bit descriptors is not interpreted as a maximum; maximum bit length and live storage are reconstructed separately. The correction and development logs are part of the release.

REFERENCES

MAURER, ANDREAS AND MASSIMILIANO PONTIL (2009): "Empirical Bernstein Bounds and Sample Variance Penalization," in *Proceedings of the 22nd Annual Conference on Learning Theory*, arXiv:0907.3740. [5]